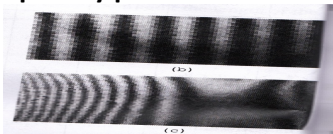
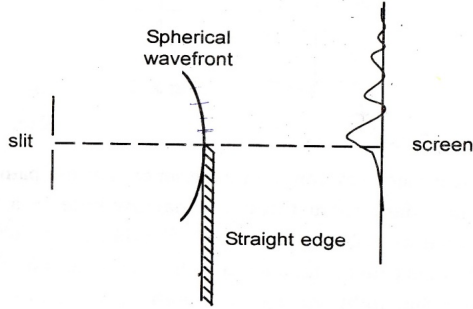
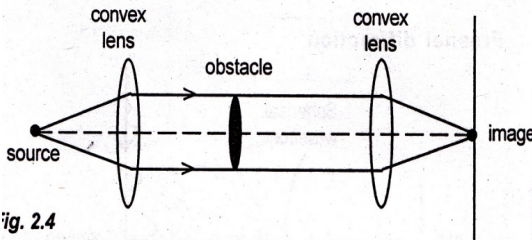
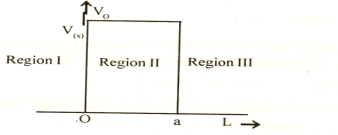
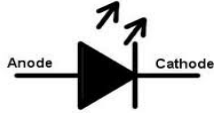


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2022		
Course Code: PHT100		
Course Name: ENGINEERING PHYSICS A		
PART A		
		Answer all questions. Each question carries 3 marks
	Q1	What do you mean by resonance in forced oscillations? Give one example
	ans	$\omega_f = \omega$ (Resonance condition) The particular frequency of the applied periodic force at which the amplitude becomes maximum is called resonant frequency. When the frequency of the applied force is equal to the natural frequency of the oscillator the amplitude will be maximum. This condition is known as resonance. example : sonometer
	Q2	Write down the three dimensional wave equation and its solution.
	ans	3 DIMENSIONAL WAVE EQUATION $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2} \text{ -----(1)}$ The solution of the above eqn is assumed as $X(x) Y(y) Z(z) = \text{const.} e^{i(k_x x + k_y y + k_z z)} \text{ -----(2)}$ The solution to the eqn is $u(x, y, z, t) = A e^{i(k_x x + k_y y + k_z z - \omega t + \phi)} \text{ -----(3)}$ Where A is amplitude and ϕ initial Phase.
	Q3	How can you test the optical planeness of a glass plate by air wedge method?
	ans	The planeness of a surface can be tested by observing the nature of fringes obtained using air wedge method. Each fringe is the locus of all points where the thickness of air film has a constant value. Since thickness of air film is same along a line parallel to the line of contact of the glass plate, the interference pattern are parallel and straight to the line of contact. If the fringes are straight and of equal thickness, the surfaces are optically plane. If the fringes are irregular (not straight) and not of equal thickness, the surfaces are not optically plane 
	Q4	Distinguish between Fresnel and Fraunhofer classes of diffraction,

ans	FRESNEL DIFFRACTION Either the source of light or the screen or both are at finite distance from the obstacle causing diffraction. Wavefront falling on the obstacle is spherical .Lenses are not used.  Example :Diffraction at a straight edge FRAUENHOFER DIFFRACTION The source of light and the screen are at infinite distance with respect to the obstacle causing diffraction. Wavefront falling on the obstacle is plane. Lenses are used  Example:Diffraction at a grating
Q5	What is quantum mechanical tunnelling? Give two examples based on this phenomenon.
ans	 Consider particle of mass m incident on the barrier from left Classically If $E < V_0$ all the particle will be reflected at the barrier If $E > V_0$ all will be transmitted in to the region III Quantum mechanically If $E < V_0$ there will be a finite probability for the particle to be transmitted in to the region III This phenomenon of tunnelling through the barrier is known as quantum mechanical tunnelling or barrier penetration . The concept is used to explain a number of phenomena in physics 1. Zener diode and Tunnel diode 2.The emission of αparticles from the radioactivenuclei 3. Barrier penetration in tunnel diode and josephson junction 4. Electron tunnelling in scanning tunnelling microscope
Q6	What are nanomaterials? Why do these materials exhibit properties different from those of their classical counterparts?
ans	The material which have at least one dimension in the order of nanometer are called

	nanostructured materials The important factors that cause significant change in the properties of nano materials are • Increased surface to volume ratio • Quantum confinement effects Quantum confinement is the restricted motion of randomly moving electron in specific energy levels when the dimensions of a material approach the de Broglie wavelength of electron Nano material have a relatively larger surface area to volume ratio when compared to bulk solids. Consider a sphere of radius 'r' . The surface area $= 4\pi r^2$ Its volume $= \frac{4}{3}\pi r^3$ Ratio $= \frac{\text{surface area}}{\text{volume}} = \frac{3}{r}$ Thus when the radius of the sphere decreases, its surface area to volume ratio increases. Hence as the particle size decreases a greater number of atoms are found at the surface compared to those inside. Thus nano material have more surface area per given volume compared with larger particles. It makes material more chemically reactive as chemical reaction occurs at surfaces.
Q7	Differentiate between Magnetic susceptibility and Magnetic permeability. Write the relation between them.
ans	MAGNETIC SUSCEPTIBILITY The magnetic susceptibility is a measure of how much a material become magnetised in an applied magnetic field. It is defined as the ratio of magnetization M to the applied magnetizing field intensity H. $\chi = \frac{M}{H}$ MAGNETIC PERMEABILITY The magnetic permeability is defined as the property of the material to allow the magnetic lines of force to pass through it. The permeability of the material is equal to the ratio of the flux density B in the medium to the field intensity H. $\mu = \frac{B}{H}$ $\mu_r = \frac{\mu}{\mu_0}$ μ_0 is the permeability of free space, $\mu_0 = 4\pi \times 10^{-7}$ Henry /meter $\frac{\mu}{\mu_0} = (1 + \chi)$ $\mu_r = 1 + \chi$
Q8	State and explain the equation of continuity for time varying fields.
ans	The total current flowing out of some volume must be equal to the rate of decrease of charge with in the volume, since the charge is conserved. In mathematical terms ,this conservation of charge concept can be stated as $I = - \frac{\partial q}{\partial t}$ $\int J \cdot ds = - \frac{\partial}{\partial t} \int \rho \, dv$

		Where ρ is the charge per unit volume or volume charge density. ($\rho=q/v$) $\int J \cdot ds = - \int \frac{\partial}{\partial t} \rho \, dv$ By using Gauss theorm, convert surface integral to volume integral $\int (\nabla \cdot J) \, dv = - \int \frac{\partial}{\partial t} \rho \, dv$ To hold this equation the integrands must be equation $\nabla \cdot J = \frac{-\partial \rho}{\partial t}$ This is the equation of continuity for time varying fields. $\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$
	Q9	Mention three advantages of fibre optic communication system.
	ans	• Extremely wide Bandwidth (Very high information carrying capacity) • Smaller diameter, lighter weight cable • Lack of cross talk between the parallel fibers. • Immunity to electromagnetic interference • Electric isolation • Low cost
	Q10	Explain the working of LED.
	ans	LED is a PN junction diode that gives off visible light when forward biased. When a LED is forward biased, the electrons from n-region and holes from p-region combine to produce energy and that will be emitted as light radiation. The electrons are in the conduction band (C.B) of n-type and at a higher state. The holes in the valence band (V.B) of p-type which are at a lower energy state. When forward biased electrons and holes recombine and the excess energy of electrons is released in the form of light.  Depending on the band gap light emitted will be in the IR region or visible region. LED band gap -1.8 eV to 2.8 eV. By doping we can modify the band gap. The frequency of the emitted light is governed by the band gap of the semiconductor. Frequency = E_g / h , where E_g is the band gap energy
PART B		
Answer any one full question from each module. Each question carries 14 marks		
Module 1		
	Q11 a	Frame and solve the differential equation of a damped harmonic oscillator. Derive the expression of displacement for underdamped, critically damped and over damped conditions and plot the results in a time - displacement graph.

ans	A harmonic oscillator in which the motion is damped by the action of an additional force is called a damped harmonic oscillator. Damped oscillations are oscillations under the action of resistive forces. Damping force is $\propto$ to its- velocity. ie Damping force = $-b \frac{dx}{dt}$ 'b' is called the damping constant. Including the damping force, the differential eq. for a damped harmonic oscillator is $m \frac{d^2 x}{dt^2} = -kx - b \frac{dx}{dt}$ $m \frac{d^2 x}{dt^2} + kx + b \frac{dx}{dt} = 0$ $\frac{d^2 x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0 \text{-----(1)}$ Let $\frac{b}{m} = 2\gamma$ where γ is the damping coefft. $\frac{d^2 x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega_o^2 x = 0 \text{-----(2)}$ This is the differential eqn for damped harmonic oscillator Solutions to the Equation Assume the solution is of the form $x = Ae^{\alpha t}$ $\frac{dx}{dt} = \alpha Ae^{\alpha t}$ $\frac{d^2 x}{dt^2} = \alpha^2 Ae^{\alpha t} = \alpha^2 x$ Putting these values in (2), we get $\alpha^2 x + 2\gamma \alpha x + \omega_o^2 x = 0$ $\text{ie } \alpha^2 + 2\gamma \alpha + \omega_o^2 = 0$ $\alpha = \frac{-2\gamma \pm \sqrt{(4\gamma^2 - 4\omega_o^2)}}{2}$ The solution
-----	---

$$x = Ae^{(-\gamma \pm \sqrt{\gamma^2 - \omega^2})t}$$

The solutions can be written as

$$x_1 = A_1 e^{(-\gamma + \sqrt{\gamma^2 - \omega^2})t} \text{ and } x_2 = A_2 e^{(-\gamma - \sqrt{\gamma^2 - \omega^2})t}$$

$\therefore$ The general solution is $x = x_1 + x_2$

$$x = A_1 e^{(-\gamma + \sqrt{\gamma^2 - \omega^2})t} + A_2 e^{(-\gamma - \sqrt{\gamma^2 - \omega^2})t} \text{ -----(3)}$$

Where A_1 and A_2 are constants

Case 1: $\gamma > \omega_0$ (high damping)

The roots of the auxiliary equation are real

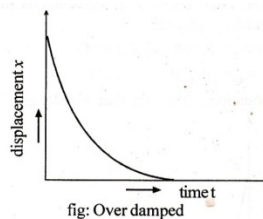
$$\alpha_1 = -\gamma + \sqrt{\gamma^2 - \omega^2}; \quad \alpha_2 = -\gamma - \sqrt{\gamma^2 - \omega^2} \text{ -----(4)}$$

The solution of equation (3) is

$$x = A_1 e^{\alpha_1 t} + A_2 e^{\alpha_2 t} \text{ -----(5)}$$

Where A_1 and A_2 are constants.

This shows that x is always positive and decreases exponentially to zero without any oscillations. The motion is called over damped . Applications- door closing mechanism



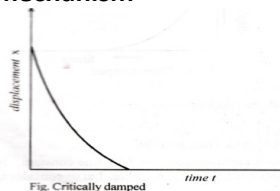
Case 2: $\gamma = \omega_0$

The roots of the auxiliary equation are real and equal each being equal to $-\gamma$.

The general solution is then

$$x = (A_1 + A_2 t) e^{-\gamma t}$$

Again x is always positive and decreases to zero. Hence this motion is termed as critically damped. Here the particle acquires the position of equilibrium very rapidly. Application- door closer mechanism



Case3: $\gamma < \omega$ The damping is low

Here $\sqrt{\gamma^2 - \omega^2}$ is imaginary

Let $\sqrt{\gamma^2 - \omega^2} = i\omega' = i\sqrt{\omega^2 - \gamma^2}$

Where $\omega' = \sqrt{\omega^2 - \gamma^2}$ -----(8)

From eq (3),

$X = A_1 e^{(-\gamma + i\omega')t} + A_2 e^{(-\gamma - i\omega')t}$ -----(9)

$= e^{-\gamma t} (A_1 e^{i\omega' t} + A_2 e^{-i\omega' t})$

$= e^{-\gamma t} (A_1 (\cos \omega' t + i \sin \omega' t) + A_2 (\cos \omega' t -$

$i \sin \omega' t)$

$X = e^{-\gamma t} ((A_1 + A_2) \cos \omega' t + i(A_1 - A_2) \sin \omega' t)$

Put $(A_1 + A_2) = x_0 \sin \varphi$

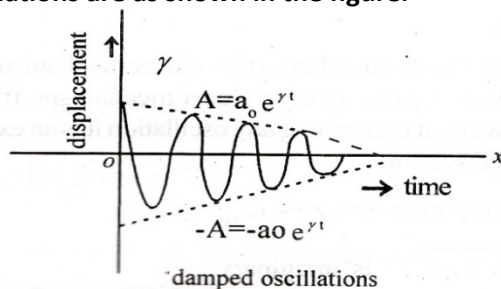
And $i(A_1 - A_2) = x_0 \cos \varphi$

$X = x_0 e^{-\gamma t} \sin(\omega' t + \varphi)$ ----- (10)

Equation (10) shows that the motion is oscillatory. The amplitude $x_0 e^{-\gamma t}$ is not a constant but decreases with time. The motion is not periodic. (under damped condition)

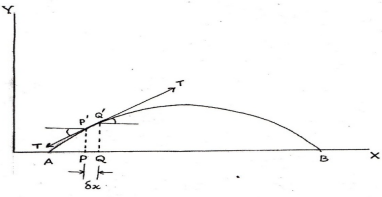
Period of oscillation $T = \frac{2\pi}{\omega'} = \frac{2\pi}{\sqrt{\omega^2 - \gamma^2}}$

The oscillations are as shown in the figure.



b

The amplitude of an underdamped harmonic oscillator reduces to 1/10th of its initial value after 100 oscillations. Its time period is 1.15s. Calculate the damping constant and relaxation time.

ans	$A = A_0 e^{-\gamma t}$ $t = 100 \times 1.15 = 115$ $\frac{A_0}{10} = A_0 e^{-115\gamma}$ $\frac{1}{10} = e^{-115\gamma}$ $e^{115\gamma} = 10$ $115\gamma = \ln_e 10$ $\gamma = \frac{2.303}{115} = 0.0200 \text{ s}^{-1}$ $\tau = \frac{1}{2\gamma} = \frac{1}{2 \times 0.0200} = 25 \text{ sec.}$
Q12 a	Derive an expression for the fundamental frequency of transverse vibrations of a stretched string.
ans	Consider an infinitely long thin and uniform string stretched between two points by a constant tension T along the X-axis. Let the string be slightly displaced aside along the Y-axis and released. Transverse vibrations are set up in the string. Consider a small element PQ of the string of length δx. The magnitude of tension in the string will be same everywhere since the string is perfectly flexible. The tension T acts tangentially at every point on the string. $P'Q'$ is the displaced position of the string.  The tension at P' and Q' acts along the tangent making angles θ_1 and θ_2 with the horizontal. Resolving the tension along x-axis and y-axis, Then net force on PQ acting X and Y direction are $F_x = T \cos \theta_2 - T \cos \theta_1$ $F_y = T \sin \theta_2 - T \sin \theta_1$ For small oscillations θ_1 and θ_2 are very small. $\cos \theta_1 = 1 \quad ; \quad \cos \theta_2 = 1$ Also $\sin \theta_1 = \tan \theta_1 \quad ; \quad \sin \theta_2 = \tan \theta_2 \quad (\tan \theta_1 = \sin \theta_1 / \cos \theta_1 = \sin \theta_1)$ Then $F_x = 0$ $F_y = T \tan \theta_2 - T \tan \theta_1$ So the net force acting on the element δx in the displaced position is only along Y-axis. $F_y = T(\tan \theta_2 - \tan \theta_1)$ If μ is the mass per unit length of the string (linear density), mass of the element δx, $m = \mu \delta x$ $\text{Acceleration} = \frac{\partial^2 y}{\partial t^2}$ According to the Newton's second law of motion $F = ma$

$$F = \mu \delta x \frac{\partial^2 y}{\partial t^2}$$

$$T (\tan \theta_2 - \tan \theta_1) = \mu \delta x \frac{\partial^2 y}{\partial t^2}$$

$$\text{But } \tan \theta_1 = \left(\frac{\partial y}{\partial x} \right)_x ; \quad \tan \theta_2 = \left(\frac{\partial y}{\partial x} \right)_{x+\delta x}$$

$$\text{or } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\tan \theta_2 - \tan \theta_1}{\delta x} \right]$$

$$\text{ie } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right]$$

when $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right] = \frac{\partial^2 y}{\partial x^2} \quad \text{--- (5)}$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2} \quad \text{--- (6)}$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2}$$

This is the differential equation of a vibrating string.

Comparing this equation with the standard wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

$$\frac{1}{v^2} = \frac{\mu}{T}$$

$$v^2 = \frac{T}{\mu}$$

$$v = \sqrt{\frac{T}{\mu}}$$

This is the velocity of transverse wave on a stretched string.

$$v = v\lambda$$

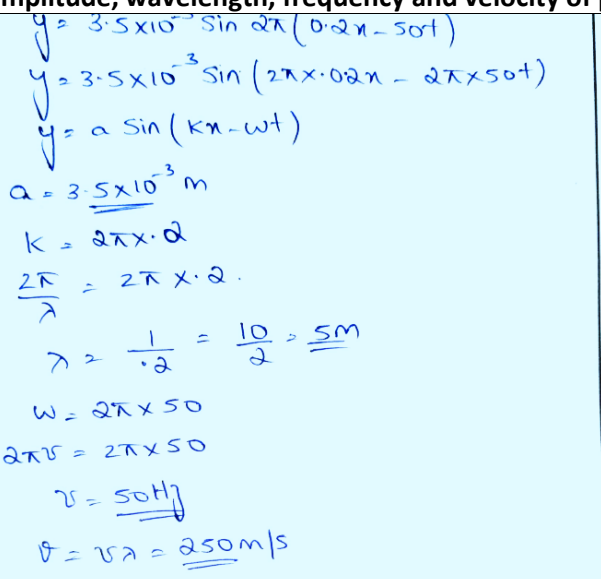
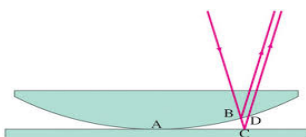
$$v = \frac{v}{\lambda}$$

$$v = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$$

This is the frequency of transverse waves developed on a stretched string.

Modes of vibrations

If the string is 'l' meters long and vibrating in one segment only ,then $l = \frac{\lambda}{2}$

	 $\lambda = 2l$ The frequency of transverse wave is given by $v = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$ Frequency of the standing wave with n antinodes is given by $v = \frac{n}{2l} \sqrt{\frac{T}{\mu}}$ Where , n is the no. of antinode <table><tr><td>$n = 1$</td><td>1st Harmonic or fundamental frequency</td></tr><tr><td>$n = 2$</td><td>2nd Harmonic</td></tr><tr><td>$n = 3$</td><td>3rd Harmonic</td></tr></table>	$n = 1$	1 st Harmonic or fundamental frequency	$n = 2$	2 nd Harmonic	$n = 3$	3 rd Harmonic
$n = 1$	1 st Harmonic or fundamental frequency						
$n = 2$	2 nd Harmonic						
$n = 3$	3 rd Harmonic						
b	The equation of a wave travelling in a string is given by $y = 3.5 \times 10^{-3} \sin 2\pi(0.2x - 50t)$ where x is measured in meters and t in seconds. Evaluate the amplitude, wavelength, frequency and velocity of propagation.						
ans							
Q13 a	Explain the formation of Newton's rings and show that the radius of dark ring is proportional to the square root of natural numbers. How can we use Newton's rings experiment to determine the refractive index of a liquid.						
ans	NEWTON'S RINGS  The arrangement consists of a plano convex lens of large radius of curvature placed on an optically plane glass plate. A thin film of air of varying thickness is formed between the lens and the glass plate . The thickness of thin film is zero at the point of						

contact and gradually increases towards the edge of the lens.

A beam of monochromatic light is incident normally on the lens. A part of the light is reflected from the top surface of the thin film and another part of light is reflected from the top surface of the glass plate. These two reflected beams interfere destructively or constructively and produce dark or bright band.

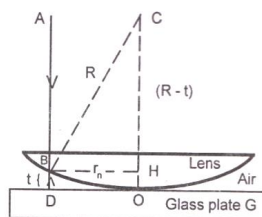
If a point appears dark, all the points along a circle through this point are dark since the thickness of air film is same along a circle. So we get a dark ring. If the point appears bright, we get a bright ring. Thus alternate dark and bright rings of increasing radii are observed. As the radii increase, the rings become thinner and closer. These are called Newton's rings.



DARK CENTRAL SPOT IN NEWTON'S RINGS

At the point of contact between the lens and the glass plate, the thickness of air film $t=0$. Hence the path difference $2\mu t \cos r = 0$. But at the time of reflection, the reflected wave from the glass plate undergoes a phase change of π (path difference of $\frac{\lambda}{2}$). So the waves at the centre are interfering destructively and hence a dark spot is obtained at the centre of Newton's ring.

RADIUS OF THE n^{th} DARK RING



Let C be the centre of curvature of the curved surface of the lens and R be its radius of curvature. Let n^{th} dark ring be formed through B where the thickness of the air film is 't'. Let r_n be the radius of dark ring, i.e. $BH=r_n$

The condition for the formation of the n^{th} dark ring at D is,

$$2 \mu t \cos r = n\lambda \dots\dots\dots(1)$$

where $n = 0, 1, 2, 3 \dots\dots\dots$

Since the light is incident normally, $r=0$ or $\cos r = 1$.

Also $\mu = 1$ for air.

$$\therefore 2t = n\lambda \dots\dots\dots(2)$$

From the right angled triangle CBH,

$$CB^2 = CH^2 + BH^2$$

$$\begin{aligned} \text{ie, } R^2 &= (R - t)^2 + r_n^2 \\ &= R^2 + t^2 - 2Rt + r_n^2. \end{aligned}$$

Since t is very small t^2 is negligible.

$$\therefore R^2 = R^2 - 2Rt + r_n^2$$

$$\text{or } r_n^2 = 2Rt \dots\dots\dots(3)$$

$$2t = n\lambda \text{ from (2).}$$

$$\therefore r_n^2 = Rn\lambda \dots\dots\dots(4)$$

$$\text{or } r_n = \sqrt{Rn\lambda} \dots\dots\dots(5)$$

Radius of the ring, $r \propto \sqrt{n}$

As n increases, the distance between the rings decreases. That is, the rings come closer as we move away from the centre.

WAVELENGTH OF LIGHT

The radius of the n^{th} dark ring is

$$r_n = \sqrt{Rn\lambda}$$

$$\text{ie } r_n^2 = Rn\lambda$$

If D_n is the diameter of the n^{th} dark ring,

$$r_n = \frac{D_n}{2}$$

$$\text{or } r_n^2 = \frac{D_n^2}{4} = R n \lambda$$

$$\text{or } D_n^2 = 4R n \lambda \dots\dots\dots(9)$$

If D_{n+k} is the diameter of the $(n+k)^{\text{th}}$ dark ring,

$$D_{n+k}^2 = 4R (n+k) \lambda \dots\dots\dots(10)$$

$$\text{Then } D_{n+k}^2 - D_n^2 = 4Rk \lambda.$$

$$\text{or } \lambda = \frac{D_{n+k}^2 - D_n^2}{4kR} \dots\dots\dots(11)$$

Thus, measuring D_{n+k} , D_n , and R , the wavelength λ can be calculated.

Similarly, the radius of the n^{th} bright ring, r_n , is given by

$$r_n^2 = R(2n-1) \frac{\lambda}{2}$$

$$\text{Then } r_n = \frac{D_n}{2}$$

$$\text{or } r_n^2 = \frac{D_n^2}{4} = R(2n-1) \frac{\lambda}{2}$$

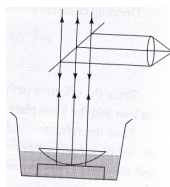
$$\text{or } D_n^2 = 4R(2n-1) \frac{\lambda}{2} \dots\dots\dots(12)$$

$$D_{n+k}^2 = 4R[2(n+k)-1] \frac{\lambda}{2} \dots\dots\dots(13)$$

$$D_{n+k}^2 - D_n^2 = 4Rk\lambda$$

$$\lambda = \frac{D_{n+k}^2 - D_n^2}{4kR}$$

REFRACTIVE INDEX OF LIQUID



Let a liquid of refractive index μ be introduced between the glass plate and the lens. The radius of the n^{th} dark ring is given by

$$r_n^2 = 2Rt$$

$$\text{or } 2t = \frac{r_n^2}{R} \dots\dots\dots(17)$$

where 't' is the thickness of the liquid film where the n^{th} dark ring is formed and R is the radius of curvature of the lens.

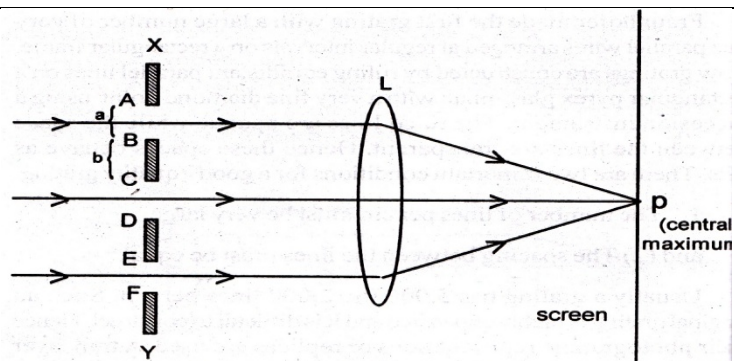
The condition for the formation of the n^{th} dark ring is

$$2\mu t \cos r = n\lambda$$

Since $\cos r = 1$ for normal incidence.

$$\text{Then } 2t = \frac{n\lambda}{\mu} \dots\dots\dots(18)$$

		$\frac{r_n^2}{R} = \frac{n \lambda}{\mu}$ $\text{or } r_n^2 = \frac{n R \lambda}{\mu} \dots\dots\dots(19)$ If d_n is the diameter of the n^{th} dark ring, $r_n = \frac{d_n}{2}$ $\therefore \frac{d_n^2}{4} = \frac{n R \lambda}{\mu}$ $\text{or } d_n^2 = \frac{4 n R \lambda}{\mu} \dots\dots\dots(20)$ If d_{n+k} is the diameter of the $(n+k)^{\text{th}}$ dark ring, $d_{n+k}^2 = \frac{4 (n+k) R \lambda}{\mu} \dots\dots\dots(21)$ $\therefore d_{n+k}^2 - d_n^2 = \frac{4 k R \lambda}{\mu} \dots\dots\dots(22)$ With air between the lens and the glass plate, we have, $D_{n+k}^2 - D_n^2 = 4 k R \lambda \dots\dots\dots(23)$ Dividing eqn (23) by eqn (22) The refractive index of the liquid, $\mu = \frac{(D_{n+k}^2 - D_n^2)}{(d_{n+k}^2 - d_n^2)}$
	b	In Newton's ring experiment the radius of the 10th dark ring is 0.75cm. When the air film is replaced by a drop of liquid, the radius reduces to 0.65cm. Find the refractive index of the liquid.
	ans	$r_{10} = 0.75 \text{ cm (in air)}$ $r_{10} = 0.65 \text{ cm (in liquid)}$ $\mu = \frac{D_{n(\text{air})}^2}{D_{n(\text{liquid})}^2} = \frac{(0.75)^2}{(0.65)^2} = \underline{\underline{1.33}}$
	Q14 a	Derive grating equation with proper diagram. What is the effect of increasing the number of lines on the dispersive power of grating?
	ans	PLANE TRANSMISSION GRATING is a plane glass plate containing a large number of equidistant parallel lines drawn using a fine diamond point . The space between the lines acts as narrow slits through which light is transmitted. The lines are opaque to light. Grating is an arrangement of a large number of parallel slits of equal width separated by equal opaque spaces . Usually a grating has 5000 to 12000 lines per cm. There are two important conditions for a good quality grating. 1. The number of lines per cm must be very large. 2. The spacing between the lines must be equal.



Consider a plane transmission grating placed perpendicular to the plane of paper. AB represents a slit and BC represents a line. Let 'a' be the width of each slit and 'b' be the width of each line. The distant (a+b) is called grating element or grating constant. A plane wavefront of monochromatic light of wavelength λ is falling normally on this slit. Each point of the wave front sends out secondary waves in all direction. The straight and parallel waves from each point can be focussed on the screen using lens. These straight waves path difference will be zero. They will interfere constructively producing brightness at the centre. This central bright band is called central maximum.

The position of central maximum is same for all the wavelength. The central maxima will have the same colour as the incident light.

Consider two waves diffracted from two points A and C of slit. They travel along AM and CN. Draw CK perpendicular to AM. Then the path difference between the two waves is AK.

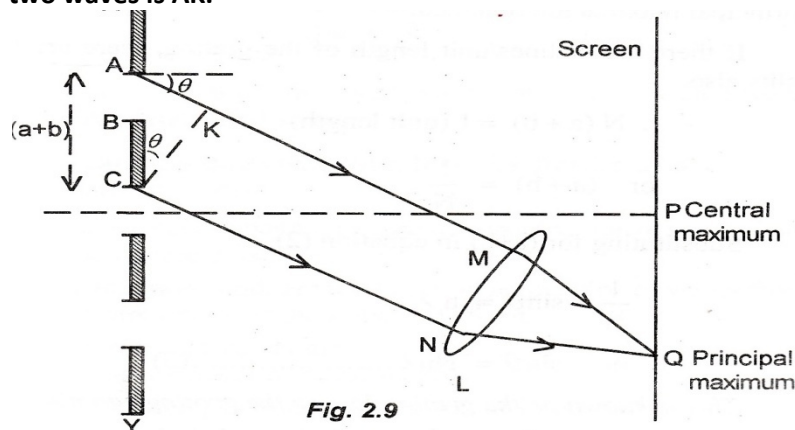


Fig. 2.9

From triangle ACK

$$\sin \theta = AK/AC$$

$$AK = AC \sin \theta$$

$$AK = (a+b) \sin \theta$$

$$\text{If } (a+b) \sin \theta = n\lambda \dots\dots(1)$$

where $n=0,1,2,3,\dots$ two waves interfere constructively. This is called principle maximum. For different values of n , there are different values of θ .

If $n=1$, it is the first order principle maximum, if $n=2$, it is the second order principal maximum and so on. Thus on either side of central maximum, a number of principal maxima are obtained.

If there are N lines/unit length of the grating, there are N slit also.

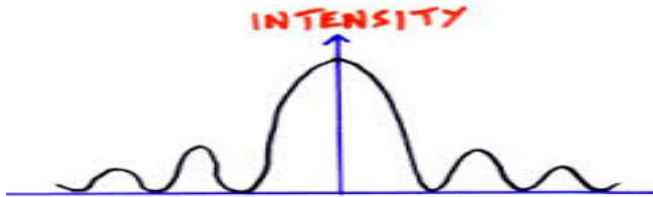
$$N(a+b) = 1 \text{ (unit length)}$$

$$(a+b) = 1/N$$

Substitute in equation (1)

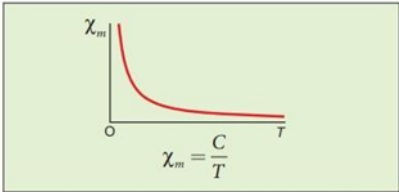
$$1/N \sin \theta = n\lambda$$

$$\sin \theta = Nn\lambda$$

		This is known as grating law or grating equation.  For a grating θ is different for different colours (λ), for each value of n. If white light is used, it gets split up into different colours. The dispersive power of a grating is defined as the ratio of the change in the angle of diffraction to the corresponding change in wavelength. Let two wavelengths λ and $\lambda + d\lambda$ be diffracted through angles θ and $\theta + d\theta$ respectively. Then the dispersive power of the grating is Dispersive Power, $\frac{d\theta}{d\lambda} = \frac{Nn}{\cos \theta}$ For small values of θ, $\cos \theta = 1$ $\frac{d\theta}{d\lambda} = Nn$ Dispersive power is proportional to the order n and the number of lines per unit length N. When Dispersive power of grating increases the number of lines per unit length of the grating also increases.
	b	At what angle will 650nm light produce a second order maximum when falling on a grating whose grating element is 1.2×10^{-3} cm.
	ans	$d \sin \theta = n \lambda$ $n = 2$ $\theta = ?$ $\lambda = 650 \times 10^{-7} \text{ cm}$ $\sin \theta = \frac{2 \times 650 \times 10^{-7}}{1.2 \times 10^{-3}}$ $= 0.1083$ $\theta = \sin^{-1}(0.1083) = \underline{\underline{6.22^\circ}}$
Module 3		
	Q15 a	What are matter waves? Obtain an expression for de Broglie wavelength. Derive expressions for the de Broglie wavelength of an electron (i) accelerated from rest through a potential of V volts (ii) having kinetic energy T .
	ans	MATTER WAVES OR DE BROGLIE WAVES Louis de Broglie introduced the concepts of matter waves. De Broglie argued that nature was symmetrical, then the fundamental quantities matter and energy should display symmetrical characters. If light shows dual nature then matter also shows particle nature and wave nature. De Broglie hypothesis says that every moving matter exhibits wave-like properties. The wave associated with a moving particle is called matter wave.

	DE BROGLIE WAVELENGTH Let c be the velocity of light. consider a photon of momentum p and frequency ν $mc^2 = h\nu$ The momentum of the photon, $p = mc = h\nu/c$ $\frac{\nu}{c} = \frac{1}{\lambda}$ $p = \frac{h}{\lambda} \quad \text{ie} \quad \lambda = \frac{h}{p}$ If m is the mass and v is the velocity of the particle, then De Broglie wavelength is $\lambda = \frac{h}{p} = \frac{h}{mv}$ This is known as de Broglie wave equation. DE BROGLIE WAVELENGTH FOR ELECTRONS Consider an electron of mass m and charge e subjected to a potential difference of V volts. If v is the velocity acquired by the electron, then $\frac{1}{2}mv^2 = eV$ De Broglie wavelength of an electron accelerated from rest through a potential of V volts $\lambda = \frac{h}{\sqrt{2meV}} = \sqrt{\frac{150}{V}} \text{Å} = \frac{12.3}{\sqrt{V}} \text{Å}$ The wavelength of electron is inversely proportional to the square root of accelerating potential. De Broglie wavelength of an electron having kinetic energy T. $\lambda = \frac{h}{\sqrt{2mT}}$
b	An electron is confined to one dimensional potential box of width 25Å. Calculate the energies corresponding to the first and second quantum states in eV.
ans	$E_n = \frac{n^2 h^2}{8mL^2}$ $L = 25 \times 10^{-10} \text{ m}$ $E_1 = \frac{1 \times (6.626 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (25 \times 10^{-10})^2}$ $= \frac{0.00096 \times 10^{-17}}{1.6 \times 10^{-19}} = 0.0602 \text{ eV}$ $E_2 = \frac{4 \times (6.626 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (25 \times 10^{-10})^2}$ $= 0.2408 \text{ eV}$
Q16 a	Explain optical, electrical and mechanical properties of nanomaterials. Write any four applications of nanomaterials in the medical field.
ans	Electrical properties • Ionization potential at nano sizes are higher than that at bulk solids.

		• The change in electrical properties is not the same in all nano materials. • In nano ceramics the electrical conductivity increases with the decrease in particle size. • But in metals it decreases. • In the case of bulk metals the energy level of electron is continuous where as in nano sized metals the energy levels are discrete. Optical properties • Depending on the particle size, same substance shows different colours. • Gold spheres of 100 nm size appears orange while that of 50 nm size appears green The particles can be made to emit or absorb specific wavelength of light by varying its size Mechanical properties • Nano crystalline nickel is as strong as hardened steel. Because of nano size many mechanical properties such as hardness, elastic modulus, fracture toughness, scratch resistance and fatigue strength are modified • 30-50 % lower elastic moduli than conventional materials • 2-7 times higher hardness • Super plastic behaviour in brittle ceramics Applications of Nanotechnology <u>Nanomedicine</u> • Cell imaging • Treating cancer • For diagnostic purposes • The drug consumption and side effects can be reduced significantly by targeted medicine. • This is done depositing the medicine in the affected region only and in the correct doze • NT helps to reproduce or repair damaged tissue.
	b	What are quantum dots and quantum sheets?
	ans	Quantum sheets Quantum sheets are two dimensional structure in which quantum confinement act only in one direction. Quantum confinement act along the thickness of the film. Electrons are free to move in xy plane of the film. So quantum confinement is in one dimension and are free to move in 2 dimensions. They are now used to make semiconductor lasers, LEDs and solar cell . Quantum dots Quantum dots are zero dimensional structure in which the electron is confined in all three dimensions. Their energy states are quantized in all three directions. consider a tiny dot of semi conductor material then we have a 3D quantum confinement of charge carriers. That is there are no freely moving electrons or holes in any direction .This is called “quantum dot”. Although this quantum dot consists of thousands of atoms it behaves like a single atom with discrete energy levels rather than many atoms. When the size of the crystal become so small that it approaches the size of exciton bohr radius, then the electron energy levels will change from continuous to discrete values. That is their is a finite separation between energy levels. The energy gap increases with the decreases in the diameter of the dot. So quantum dot can emit any colour from blue to red

		depending on the size of the dot.
Module 4		
Q17 a		How paramagnetic substances differ from Ferromagnetic substances? Write two examples for each of them.
ans		PARAMAGNETISM • Paramagnetic materials are weakly magnetised in an externally applied magnetic field. • Paramagnetic material show tendency to move from weak to strong magnetic field when placed in a non uniform magnetic field. • Paramagnetism is due to the presence of unpaired electrons in atoms. • Electrons undergoes spin motion and orbital motion and produces spin magnetic moment and orbital magnetic moment. • These moments are randomly oriented and exhibit no magnetic moment. When magnetic field is applied, • The electrons spins tends to align with the applied field resulting in magnetic moment • Induced field will be in the direction of applied field;but weak (weak attraction) • Free electron magnetism is weak and is called pauliPara magnetism • They do not retain any magnetism in the absence of an external magnetic field. • Susceptibility is low and positive. • Paramagnetic susceptibility depends on temperature Examples: aluminium , chromium, calcium, platinum, O₂, Ne CURIE'S LAW According to curies law, the susceptibility of a paramagnetic material is inversely proportional to its temperature. $\chi \propto \frac{1}{T}$ $\chi = \frac{C}{T}$ C is called curie's constant.  FERROMAGNETISM

		• Their magnetic property not only because of their atomic magnetic moment but also due to certain region called magnetic domain • In a domain very large number of atoms are aligned parallel to each other. • Magnetic force within the domain is very strong. • Before magnetisation the domains are randomly oriented relative to each other (net dipole moment zero). <u>When magnetic field is applied,</u> • Domains within the material are aligned • When all domains are aligned the material is said to be magnetically saturated • Strongly attracted by external field • Magnetism persists even after the removal of magnetic field. • They show hysteresis. • Loss their magnetic properties if they are heated to high temperatures. • They used to make permanent magnet. • They have high susceptibility and relatively high permeability. • Examples: iron, nickel, cobalt, magnetite
	b	State Faraday's laws of Electromagnetic induction. What is Lenz's law?
	ans	FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION Whenever the magnetic flux linked with a circuit changes an emf is induced in that circuit. According to Faraday's Law, The magnitude of induced emf is equal to the rate of change of magnetic flux through the circuit. $\text{Induced emf, } e = \frac{d\Phi}{dt}$ $e = - \frac{d\Phi}{dt} \text{ (Lenz's law)}$ -ve sign indicates the direction of the induced emf which tends to oppose the change in the magnetic flux.
	Q18 a	From the basic laws of electricity and magnetism, deduce Maxwell's four field equations.
	ans	MAXWELL EQUATIONS $\nabla \cdot \mathbf{D} = \rho$ $\nabla \cdot \mathbf{B} = 0$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Derivation of 1st equation

According to Gauss's law in electrostatics, The total normal electric flux through a closed surface is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed by the surface.

$$\text{That is } \phi_E = \frac{q}{\epsilon_0}$$

$$\int \mathbf{E} \cdot d\mathbf{s} = \frac{q}{\epsilon_0}$$

$$\int \mathbf{E} \cdot d\mathbf{s} = \frac{1}{\epsilon_0} \int \rho dv$$

$$\int \epsilon_0 \mathbf{E} \cdot d\mathbf{s} = \int \rho dv$$

$\epsilon_0 \mathbf{E} = \mathbf{D}$ is called electric displacement vector

$$\int \mathbf{D} \cdot d\mathbf{s} = \int \rho dv$$

Applying Gauss theorem

$$\int \nabla \cdot \mathbf{D} dv = \int \rho dv$$

To hold this equation the integrands must be equal.

$$\nabla \cdot \mathbf{D} = \rho \dots \dots \dots \text{Maxwell's First equation}$$

Maxwell's Second Equation

By Gauss's law in magnetism the net magnetic flux emerging through any closed surface is zero.

$$\text{Then } \int \mathbf{B} \cdot d\mathbf{s} = 0$$

By using Gauss theorem,

$$\int \nabla \cdot \mathbf{B} dv = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

For a magnetic dipole, in any closed surface the magnetic flux directed inwards towards South Pole will be equal to the flux outward from North Pole. The net flux will be zero for a dipole.

Maxwell's Third Equation

According to Faradays law of electromagnetic Induction,

Induced emf $e = -\frac{\partial \phi}{\partial t}$ ϕ is magnetic flux

$$e = -\frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{s} \dots \dots \dots (1)$$

We know that $e = \int \mathbf{E} \cdot d\mathbf{l} \dots \dots \dots (2)$

$$\int \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{s}$$

Using Stokes theorem

$$\int \nabla \times \mathbf{E} \cdot d\mathbf{s} = -\frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{s}$$

To hold this equation Integrands must be equal

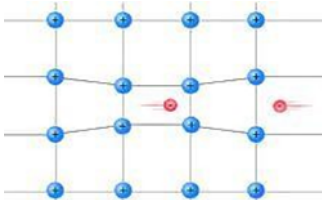
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \dots \dots \dots \text{Maxwell's third equation}$$

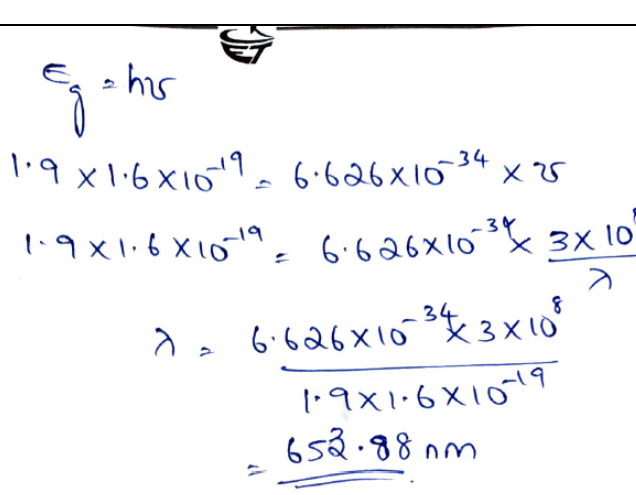
Maxwell's Fourth Equation

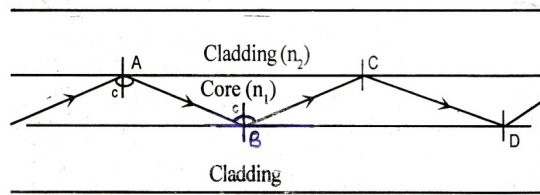
According to amperes circuital law the magnetic induction around a closed current carrying loop is equal to μ_0 times the current flowing in the loop.

$$\text{Thus } \int \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

	$\oint \mu_0 H. dl = \mu_0 I$ $\oint H. dl = I \dots\dots\dots(1)$ If J is the current density of the loop, then current $I = \int J. ds \dots\dots\dots(2)$ From eqn 1 and 2 $\oint H. dl = \int J. ds$ Using Stokes theorm $\int \nabla \times H. ds = \int J. ds$ $\nabla \times H = J \dots\dots\dots(4)$ Take divergent of eqn (4) $\nabla. \nabla \times H = \nabla. J$ But $\nabla. \nabla \times H = 0$ Since divergent of curl of any vector will be zero. Hence $\nabla. J = 0 \dots\dots\dots(5)$ From eqn of continuity $\nabla. J = -\frac{\partial \rho}{\partial t} \dots\dots\dots(6)$ Here there is incompatibility in eqn 5 and 6 Hence Maxwell introduced a new term J_D called displacement current density in eqn no (4) $\nabla \times H = J + J_D \dots\dots\dots(7)$ Now take divergent on both sides of eqn (7) $\nabla. \nabla \times H = \nabla. J + \nabla. J_D$ $0 = \nabla. J + \nabla. J_D$ $-\nabla. J_D = \nabla. J$ $\nabla. J = -\nabla. J_D$ From eqn 6 $-\frac{\partial \rho}{\partial t} = -\nabla. J_D$ From Maxwell's first law $\rho = \nabla. D$ $\frac{\partial}{\partial t} \nabla. D = \nabla. J_D$ $\nabla. \frac{\partial D}{\partial t} = \nabla. J_D$ $\frac{\partial D}{\partial t} = J_D$ now substitute this value in eqn 7 $\nabla \times H = J + \frac{\partial D}{\partial t} \dots\dots\dots \text{This is Maxwell's fourth equation}$
b	Compare displacement current and conduction current
ans	- Conduction current is produced due to rate of change of charges. But the displacement current is produced due to rate of change of Electric Field. - Conduction current is zero in perfect vacuum but displacement current has a finite value in vacuum. - Conduction current is in phase with the Electric Field whereas Displacement current leads the Electric Field in phase by $\frac{\pi}{2}$ radians. - Conduction current obeys Ohm's Law but displacement current doesn't obey Ohm's Law. - Conduction current density is represented by $J_c = \sigma E$ whereas displacement current density

		is given by $J_d = \frac{\partial D}{\partial t} = \epsilon \frac{\partial E}{\partial t}$. - Conduction current is the actual current whereas displacement current is the apparent current produced by time varying electric field. - Conduction current is dominant in conductors but displacement current is dominant in dielectrics
Module 5		
Q19	a	Explain BCS theory of superconductivity. Describe high temperature superconductors. Write three applications of superconductors.
	ans	BCS THEORY Bardeen ,Cooper and Schrieffer developed a new theory to explain superconductivity of materials and this is known as BCS theory. According to this theory,superconductivity is mainly due to the interaction of phonons and electrons  When an electron is advancing through a path in between 2 sets of positive ions in the lattice, electron attracts nearby positive ions and lattice gets distorted. This distortion produces a region of increased positive charge density and is called phonon. If a second electron approaches this region, this electron also attracted by the positive region. Thus these 2 electrons are held together by force of attraction by phonon. Thus 2 electrons are coupled together at low temperature and this pair is called cooper pair. Cooper pair behaves like single particle. Cooper pair has zero spin and zero linear momentum. At absolute zero almost all electrons are coupled together as cooper pairs. Superconducting energy gap The net energy of electrons in a cooper pair is less than the total energy of these individual electrons. There is an energy gap in the energy spectrum of electrons. All the Cooper pairs are occupying lower energy states and energy gap prevents them from unpairing. The minimum energy required to break up cooper pairs is called superconducting energy gap. When temperature increases ,energy gap decreases. At T_c energy gap becomes zero and cooper pairs are broken up and thus electrons are separated. The discovery of a new type of superconductors is one of the most important scientific events in our century. Scientists made a lot of research works to produce superconductor with high T_c. In 1977 a high transition temperature 23 K was achieved using metallic compound of niobium and Germanium .But in 1986Bedddnorz and Muller discovered La – Ba-CuO system of ceramic super conductors with a T_c of 34 K. Such superconductors with high critical temperature are called High Temperature Superconductors High Temperature Super conductors are ceramic super conductors with a high transition temperature greater than 40 K. In low temperature superconductors it is very difficult and expensive to maintain low temperature for a very long period .It is very difficult to maintain the liquid helium temperature (4.2K). nowadays it is possible to replace the expensive liquid He with cheaper and more efficient coolant liquid nitrogen. The temperature T_c can be raised using microwaves or laser radiations.

		In ceramic superconductors , the cell contains 1 atom of rare earth metal, 2 Barium atoms, 3 copper atoms and 7 Oxygen atoms. Since no. of atoms of metal are in the ratio 1:2:3 , such ceramic super conductors are called 1-2-3 super conductors and it is working at T_c of 90K. Copper oxide superconductors belongs to another group of HTSC working at T_c of 120K . The Resonating Valence Bond State Theory explained HTSC. Superconductors has a wide range of applications 1. Superconductors are used to produce a very strong and powerful magnetic field in the order of 20T at a low cost. - To bend and guide the charged particles in particle accelerators , cyclotrons etc. - Used for controlling and focusing high temperature plasma for the controlled nuclear fusion . - Used in magnetic levitation –Maglev. - Superconducting magnets are used to produce very efficient ore separating machines. 2) Medical field - MRI: The most important medical application of superconductivity is MRI. MRI is free from harmful effects of other radiations like X-rays. - Superconducting magnetic field is used to remove tumour cells from the healthy cells - A group of squids is used for the diagnosis of epilepsy - Using superconducting susceptometer the iron content in the body can be estimated . - Squids are used to measure minute magnetic fields produced from heart and brain very accurately and used for pathological analysis of their functions.
	b	A light emitting diode is made of GaAsP having a band gap of 1.9eV. Determine the wavelength of the radiation emitted.
	ans	 $E_g = \frac{hc}{\lambda}$ $\lambda = \frac{hc}{E_g}$ $\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.9 \times 1.6 \times 10^{-19}}$ $\lambda = 652.88 \text{ nm}$
	Q20 a	Explain how light is propagated through an optical fibre. Define numerical aperture of an optical fibre and derive the expression for numerical aperture of a step index fibre.
	ans	PROPAGATION OF LIGHT WAVES THROUGH AN OPTIC FIBRE



When a ray of light is incident from core to the cladding at an angle of incidence i greater than the critical angle at A, it undergoes total internal reflection at the core - cladding interface. So the ray rebounds along AB. It is again incident at B at an angle greater than the critical angle on the core - cladding interface. So it again undergoes total internal reflection and the ray is travelling along BC. The ray is undergoing successive total internal reflections at points on the core - cladding interface. Thus the wave is propagated through the optical fibre in a zig-zag manner by the multiple total internal reflection principle without any loss of energy.

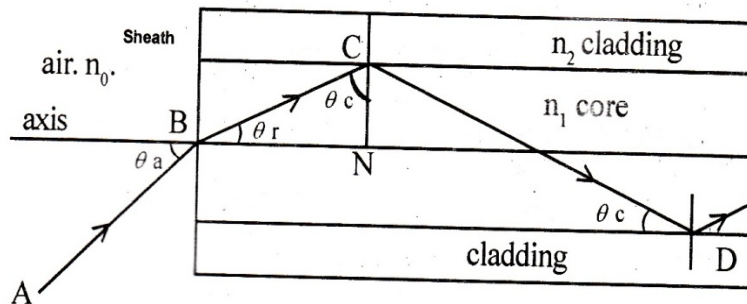
Numerical aperture NA of an optic fibre is the sine of acceptance angle θ_a .

$$NA = \sin \theta_a$$

NA depends on the acceptance angle. If θ_a is larger, NA will also be higher. Numerical aperture is also the light gathering power of the fibre and it measures the amount of light accepted by the fibre. NA is between 0.13 and 0.5. If NA is larger, fibre can accept more light from the source. NA depends on the refractive indices of core and cladding.

$$NA = \sqrt{n_1^2 - n_2^2}$$

Consider a light ray AB incident at B at the edge of the core of an optic fibre from air. It is incident at an angle θ_a with the axis of the core. This maximum angle is known as acceptance angle. Since it travels from air to core, it is refracted along BC at an angle θ_r . This refracted ray is now incident at C at the core-cladding interface with an angle slightly greater than the critical angle θ_c . Hence the ray is undergoing total internal reflection and it is travelling along CD. Thus the ray is propagated through the fibre by multiple total internal reflections.



At C, CN is drawn normal to the axis. The angle at C is taken as the limiting angle θ_c , the critical angle. Let n_0 be the refractive index of air, n_1 that of core and n_2 that of cladding.

By Snell's law at B,
$$\frac{\sin \theta_a}{\sin \theta_r} = \frac{n_1}{n_0}$$

$$n_0 \sin \theta_a = n_1 \sin \theta_r$$

$$n_0 \text{ for air} = 1$$

$$\sin \theta_a = n_1 \sin \theta_r$$

$$NA = \sin \theta_a$$

$$NA = n_1 \sin \theta_r$$

		$= n_1 \cos (90 - \theta_r)$ $\theta_c + \theta_r = 90^\circ$ $\theta_c = 90^\circ - \theta_r$ $NA = n_1 \cos (\theta_c)$ $\cos^2 \theta_c + \sin^2 \theta_c = 1$ $\cos^2 \theta_c = 1 - \sin^2 \theta_c$ $\cos \theta_c = \sqrt{1 - \sin^2 \theta_c}$ $NA = n_1 \sqrt{1 - \sin^2 \theta_c}$ At the critical angle, $\frac{\sin \theta_c}{\sin 90} = \frac{n_2}{n_1}$ $\sin \theta_c = \frac{n_2}{n_1}$ $NA = n_1 \sqrt{1 - \frac{n_2^2}{n_1^2}}$ $NA = n_1 \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$ Numerical Aperture , $NA = \sqrt{n_1^2 - n_2^2}$ Where, n_1 = Refractive index of core n_2 = Refractive index of cladding
b		In an optical fibre, the core material has refractive index 1.43 and refractive index of the cladding material is 1.4. Find numerical aperture and acceptance angle
ans		$n_1 = 1.43 \quad n_2 = 1.4$ $NA = \sqrt{n_1^2 - n_2^2}$ $= \sqrt{1.43^2 - 1.4^2} = \sqrt{0.0849}$ $= 0.2914$ Acceptance angle $= \sin^{-1}(NA)$ $= \sin^{-1}(0.2914)$ $= 16.94^\circ$

